# An Hung Nguyen

**COMPUTER SCIENTIST / AI RESEARCHER**

Danang, Vietnam

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Google Scholar: An Hung Nguyen

## PERSONAL PROFILE

AI Engineer with 5+ years of experience in Computer Vision, bridging academic research and real-world applications. Experienced in building end-to-end machine learning pipelines, from data analysis and model training to optimization and deployment on edge devices. I also have experience working with Large Language Models (LLMs) and helping customers improve the reliability, scalability, and cost-efficiency of AI solutions.

I have authored and co-authored 10+ research papers and received several academic awards throughout high school and university. Passionate about research and continuous learning, I enjoy exploring new technologies, tackling challenging problems, and turning innovative ideas into practical solutions.

## EDUCATION

**University of Science and Technology - The University of Danang** *Danang, VN*
***B.S., Information Technology*** 2021 - 2025

- Grade: 3.65/4.0
- Excellent Bachelor's Degree
- Working as an AI Researcher with Minh Tuan Pham PhD.
- Capstone Project: A Deep Learning-Based Approach for Multi-Weather Image Restoration and Enhancement. (http://thuvienso.dut.udn.vn/handle/DUT/16775)
- Courses: Data Science, Programming for Data Science, Knowledge Representation and Reasoning, Statistical Theory and Methods, Learning Skills through Case Studies, Artificial Intelligence, Machine Learning, Statistical Learning

**Le Quy Don upper-secondary school for the gifted** *Quang Tri, VN*
***Informatics class*** (***IT 18-21***) 2018 - 2021

- Specialised in Biology, Informatics, and Mathematics with Computer Science
- Selected as a member of the national-level scientific research competition team and the Informatics gifted student team.

## WORK EXPERIENCE

**SLA Software Logistik Artland GmbH** *Germany (Remote)*
***AI Software Engineer*** Aug 2025 - Sept 2026

- Developed and deployed automation systems for low-code AI model training and data management as part of the **Dubai-Premiere: TrAInity – AI-Empowerment** project.
- Built an end-to-end automated pipeline for image classification and object detection, covering dataset analysis (EDA), preprocessing, model training, and evaluation. Simplified the workflow so users only need to provide datasets and select the number of GPUs, with automated training reports generated upon completion.

**SLA Software Logistik Artland GmbH** *Germany (Remote)*
***Computer Scientist & AI Engineer*** Aug 2024 - Sept 2026

- Developed an OCR module for real-time recognition and tracking of slaughter numbers printed on animal hides, supporting automated traceability (One of them: https://lnkd.in/p/gufFDTc6).
- Supported the integration and deployment of AI models into existing customer infrastructures, including hardware selection, automated installation scripts, and system monitoring and troubleshooting.
- Developed end-to-end conversion and optimization scripts to convert PyTorch and TensorFlow models into NPU-compatible formats for efficient inference on edge devices.
- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.
- **Technical Skills**: Python with PyTorch, NumPy, Matplotlib, Pandas, Scikit-learn, C++, Ubuntu Linux, Linux tools, Apt, Scripting, Git.
- **Soft Skills**: Teamwork, Time Management, Communication, Presentation skills.
- **Research Skills**: Medical Research, Data Analysis, Data Engineering.

**University of Regina** *Canada*
***Collaboration Researcher*** Aug 2024 - Sept 2025

- Researched solutions for simulating chemical processes.
- Implemented simulation code using Matlab and Python.

**Phygital Labs** *Vietnam*
***AI Intern*** May 2024 - Jul 2024

- Worked as an AI intern on deep learning projects.
- Worked with LLM models to propose solutions for querying large documents.
- Collaborated with a former Senior Google engineer to deploy systems.

**FX Data** *Germany (Remote)*
***Computer Scientist & AI Engineer*** Jan 2022 - Jul 2024

- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.
- **Technical Skills**: Python with PyTorch, NumPy, Matplotlib, Pandas, Scikit-learn, C++, Ubuntu Linux, Linux tools, Apt, Scripting, Git.
- **Soft Skills**: Teamwork, Time Management, Communication, Presentation skills.
- **Research Skills**: Medical Research, Data Analysis, Data Engineering.

**Rhen Consulting** *Germany (Remote)*
***AI Engineer*** Oct 2021 - Dec 2021

- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.

## HIGHLIGHT AWARDS

**Fourth Place Award in the Field of Science, Engineering, and Technology** *Vietnam*
*Vietnam Student Science Research and Innovation 2022 - 2023* Dec 2023

Project title: Quaternion Convolutional Neural Networks for Image Depth Estimation

**Fourth Place Award in the in Computational Biology and Bioinformatics** *Vietnam*
*Vietnam National Secondary Student Science and Engineering Fair* March 2021

Project title: Applying Deep Learning to the process of diagnosing, analyzing and detecting damage to the retina due to diabetes

**Third Prize in the Provincial Contest for Excellent Students in Informatics** *Vietnam*
*Quang Tri Department of Education and Training* Jan 2020

## PUBLICATIONS

**[1] All-in-One Image Restoration with Quaternion Cross-Attention and Haar Frequency Decomposition**
*Submitted to Signal, Image and Video Processing (Journal Q2)* 2025

Status: Under review
**An Hung Nguyen**, Minh Tuan Pham
dx.doi.org/10.21203/rs.3.rs-7260641/v1
This work is part of my undergraduate research project and capstone thesis.

**[2] Optimizing Retrieval in Vietnamese Literary QA Using Named-Entity Relationship Graphs**
*Submitted to ACM Transactions on Asian and Low-Resource Language Information Processing (Q2)* 2025

Status: Under review
Minh Tuan Pham, **An Hung Nguyen**, Phuc Hao Do, Thanh Huy Nguyen, and Nam Do

**[3] Real-Time 3D Reconstruction through Image Synthesis and IoT Integration**
*Manuscript under review* 2025

Cong Thang PHAM, **An Hung Nguyen**, Tan Sang Truong, Dai Quy Tran, Si Tien Tran
Submitted to Vestnik St. Petersburg University: Mathematics

**[4] Adaptive Hybrid Transformer Network with Frequency Attention for All-in-One Image Restoration**
*PSBB25 — Moscow, RUSSIA* June 09 - 11, 2025

Cong Thang Pham, **An Hung Nguyen**, Quoc Cuong Nguyen, Minh Nhat Phan, Thanh Than Nguyen
10.5194/isprs-archives-XLVIII-2-W9-2025-213-2025

**[5] DehazeCLNet: A Contrastive Learning Framework with Advanced Feature Extraction for Image Dehazing**
*SOICT 2024 (Springer CCIS) — Danang, Vietnam* December 13-15, 2024

Cong Thang Pham, **An Hung Nguyen**, Quoc Cuong Nguyen, Minh Nhat Phan
10.1007/978-981-96-4282-3_23

**[6] Enhancing Image Classification Capabilities In The Vision Transformer Network Model With Quaternion Algebra**
*CITA 2024 — Danang, Vietnam* July, 2024

Minh Tuan Pham, **An Hung Nguyen**
https://elib.vku.udn.vn/handle/123456789/4031
https://nguyenhung.me/paper/QViT.pdf

**[7] Enhancing Convolutional Neural Network Architectures with Long Short-Term Memory for Improved Image Classification**
*IEEE: Institute of Electrical and Electronics Engineers — Danang, Vietnam* November 10-11, 2023

**An Hung Nguyen**, Minh Tuan Pham
10.1109/ATiGB59969.2023.10364368

**[8] Quaternion Convolutional Neural Networks for Depth Estimation**
*IEEE: Institute of Electrical and Electronics Engineers — Mie, Japan* January 4, 2023

**An Hung Nguyen**, Cao Duy Hoang, Dang Hoang Phu Phan, Minh Tuan Pham
https://doi.org/10.1109/SCISISIS55246.2022.10002074

[9] **Weight Initialization for Complex Number Convolutional Neural Networks Using Genetic Algorithms**
*Journal on Information Technologies and Communications — Vietnam* June 30, 2022
Minh Tuan Pham, **An Hung Nguyen**
https://doi.org/10.32913/mic-ict-research-vn.v2022.n1.1054

[10] **Efficient Complex Valued Neural Network with Fourier Transform on Image Denoising**
*ACM: Association for Computing Machinery — Dubai, UAE* December 16, 2021
Minh Tuan Pham, Van Quang Nguyen, Cao Duy Hoang, **An Hung Nguyen**
https://doi.org/10.1145/3508072.3508081

## SKILLS

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|---|---|
| **Programming:** | C/C++, Python (Pandas, PyTorch, NumPy, Scikit-learn. etc), Java, dot NET, React Native, SQL. |
| **Miscellaneous:** | Linux, Shell (Bash/Zsh), LATEX(Overleaf/R Markdown), Microsoft Office, Git, Arduino, MCU microcontroller. |
| **Soft Skills:** | Time Management, Teamwork, Leadership, Problem-solving, Documentation, Engaging Presentation. |

## TOP HONORS AND AWARDS

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| Mar 2025 | 2nd prizes, Faculty level Student Scientific Research and Innovation | Vietnam |
| Dec 2023 | Third prize, University-level Student Scientific Research and Innovation | Vietnam |
| Dec 2023 | First prize, Faculty level Student Scientific Research and Innovation | Vietnam |
| Dec 2023 | 4th prizes, Vietnam Student Science Research and Innovation | Vietnam |
| Jun 2023 | 3rd prizes, Faculty level Student Scientific Research and Innovation | Vietnam |
| Mar 2022 | 2nd prizes, Faculty level Student Science Research and Innovation | Vietnam |
| Aug 2021 | Second prize (ranked 1st), Innovative Generation 2021 | Vietnam |
| Mar 2021 | 4th prizes, Vietnam National Secondary Student Science and Engineering Fair | Vietnam |
| Jan 2021 | 4th prizes, Provincial Contest for Excellent Students in Informatics | Vietnam |
| Jan 2020 | 3rd prizes, Provincial Contest for Excellent Students in Informatics | Vietnam |

## LANGUAGES

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| English | Professional Proficiency (B2) |
| Vietnamese | Native proficiency |